

CANONICAL RESONANT KOOPMAN SPECTRAL FLOW FOR SCALABLE AND STABILIZED LANGUAGE MODEL REASONING

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ABSTRACT

Autoregressive large language models (LLMs) during long-horizon reasoning suffer from trajectory drift and compounding error accumulation due to unconstrained readout perturbations. Continuous-time dynamical controllers offer a promising alternative via continuous latent memory states, but existing frameworks incur a severe cubic bottleneck $\mathcal{O}(d_z^3)$ from dense matrix exponentials and uncontrolled modal mixing, restricting observable dimension and throughput. We introduce **Canonical Resonant Koopman Spectral Flow (ResoKoopman)**, a linear-complexity stabilization operator that parameterizes the lifted Koopman generator in its canonical real normal form via $d_z/2$ decoupled 2×2 blocks. This enables exact closed-form inner-time integration in $\mathcal{O}(d_z)$ time—no matrix exponentiation or numerical ODE solvers—scaling d_z up to 512 without throughput loss, while disentangling conserved long-horizon premises from fast-damped calculation noise and regulating steering via a dual spectral demand-supply gate. On the full GSM8K benchmark with a frozen Llama-3.2-3B base, ResoKoopman reaches 50.87% greedy@1 (+0.99 over Thought-in-Flight) and 61.87% maj@8 (+3.18), without touching base parameters, at up to $47\times$ lower controller latency than dense Koopman generators and $2.3\times$ faster than continuous latent-CoT baselines (Coconut, CODI). Controllers trained only on GSM8K transfer zero-shot (+9.80% greedy@1 on SVAMP, +2.00% maj@8 on MATH-500), recover +19.80% under numerical distractors, and give McNemar-significant gains ($p < 0.05$); a 2×2 factorial further shows dynamic gating becomes decisive (+2.20%) once the base co-adapts.

1 INTRODUCTION

Autoregressive large language models (LLMs) have achieved remarkable milestones on complex, multi-step quantitative and symbolic reasoning tasks Wei et al. (2022); Hendrycks et al. (2021); Lewkowycz et al. (2022). Standard Chain-of-Thought (CoT) prompting enables autoregressive models to solve hard benchmark problems by decomposing complex prompts into intermediate reasoning steps Wang et al. (2022); Yu et al. (2023); Shao et al. (2024). However, during long-horizon reasoning rollouts, autoregressive decoding inherently suffers from trajectory drift and compounding error accumulation Zou et al. (2026); Huang et al. (2025). Unconstrained readout perturbations or subtle errors in early generation steps propagate through the context window, causing catastrophic divergence from the correct reasoning trajectory over extended step counts Zelikman et al. (2022); Shao et al. (2025).

To address trajectory instability, continuous-time dynamical controllers and continuous latent representations have emerged as a promising computational paradigm Wang & Yu (2021); Patro & Agneewaran (2024). Rather than relying solely on discrete token generation, continuous latent controllers augment base transformer readouts with persistent internal state dynamics that evolve over an inner-time interval Zhou et al. (2024); Yang et al. (2026). Despite their theoretical appeal, existing continuous-time dynamical frameworks encounter three severe bottlenecks: (i) a *cubic computational cost*—Koopman or dense linear generators require matrix exponentials $\exp(\mathcal{K})$ or high-order Taylor approximations, incurring $\mathcal{O}(d_z^3)$ complexity in the observable dimension d_z Turan et al. (2025); Yao et al. (2026); (ii) *uncontrolled modal mixing*—without structural constraints,

high-frequency calculation noise pollutes conserved logical premises and induces trajectory collapse in deep rollouts Zhang et al. (2025); Gu & Dao (2023); and (iii) *solver latency*—iterative numerical ODE integrators (e.g., Runge–Kutta) impose significant per-token latency, restricting throughput Zhou et al. (2024).

In this paper, we introduce **Canonical Resonant Koopman Spectral Flow (ResoKoopman)**, a linear-complexity, spectrally decoupled dynamical stabilization operator for autoregressive language models. ResoKoopman parameterizes the lifted Koopman generator $\mathcal{K}(\phi_t) \in \mathbb{R}^{d_z \times d_z}$ directly in its canonical real normal form using $d_z/2$ independent 2×2 canonical block matrices:

$$\mathbf{K}_k(\phi_t) = \begin{bmatrix} -\gamma_k(\phi_t) & -\omega_k(\phi_t) \\ \omega_k(\phi_t) & -\gamma_k(\phi_t) \end{bmatrix}, \quad (1)$$

where $\gamma_k(\phi_t) = \text{softplus}(W_\gamma^{(k)} \phi_t) \geq 0$ represents the modal dissipation rate and $\omega_k(\phi_t) = W_\omega^{(k)} \phi_t$ represents the angular phase frequency, both conditioned on the compressed readout feature ϕ_t .

This canonical parameterization provides three core benefits: (i) **exact closed-form $\mathcal{O}(d_z)$ integration**—inner-time dynamics over an interval τ are computed analytically via elementwise block-rotational exponentials $\exp(\tau \mathbf{K}_k) = e^{-\tau \gamma_k} \mathbf{R}(\tau \omega_k)$ (Section 3), eliminating numerical integrators, matrix exponentials, and Taylor approximations; (ii) **spectral disentanglement**—non-negative damping $\gamma_k \geq 0$ separates persistent long-horizon premises ($\gamma_k \rightarrow 0$) from fast-damped calculation noise (high γ_k), mitigating modal mixing; and (iii) **dual spectral demand–supply gating**—a phase-desynchronization demand gate $\mathcal{D}_t^{\text{phase}}$ and a conserved-energy supply gate $\mathcal{S}_t^{\text{energy}}$ regulate steering with do-no-harm behavior in confident regimes.

On the full GSM8K benchmark with a frozen Llama-3.2-3B base, ResoKoopman achieves **50.87% greedy@1** and **61.87% maj@8**, surpassing the continuous controller Thought-in-Flight (49.89%/58.68%) and the frozen base (41.62%/56.71%) without base fine-tuning, scales smoothly to $d_z = 512$ with near-flat overhead (up to $47\times$ faster than dense Koopman generators), and runs $2.3\times$ faster than continuous latent-CoT baselines (Coconut, CODI).

2 RELATED WORK

2.1 TEST-TIME STEERING AND REPRESENTATION CONTROL

Extending the multi-step problem-solving capabilities of large language models at inference time has driven significant research in prompt engineering and decoding-time interventions Wei et al. (2022); Zhou et al. (2024). One line of work introduces discrete computation injection, embedding learnable dummy or pause tokens into the input stream to afford the transformer extra computational steps before generating textual answers Goyal et al. (2023); Zelikman et al. (2024); Begin et al. (2025). Although pause-based token insertion helps redistribute internal attention maps, discrete token padding lacks persistent continuous memory states, making inference sensitive to sequence length variations and susceptible to context position degradation Liu et al. (2023).

In parallel, representation control techniques modify intermediate hidden states along steering vectors extracted from model activations Li et al. (2023); Rinsky et al. (2023), and have been adapted to preference alignment Cava & Tagarelli (2025), latent multi-hop path activation Yang et al. (2024); Tang et al. (2025), sparse feature control Xin et al. (2025), and targeted task modulation Højer et al. (2025). A closely related line replaces explicit token rationales with *continuous latent Chain-of-Thought*: Coconut Hao et al. (2024) recurrently feeds the last hidden state back as the next input embedding, and CODI Shen et al. (2025) distills such latent thoughts into a compact head Shao et al. (2025), alongside agentic inference-time steering Xia et al. (2026) and associative spectral memory Sridhar & Johansen (2026). Crucially, both Coconut and CODI require multi-stage *fine-tuning of the entire base Transformer* and insert N_{latent} additional sequential forward passes per thought, roughly $2.3\times$ the base per-token latency (Section 4.7). Moreover, reasoning within unconstrained hidden spaces is unstable: un-gated or static residual steering lacks continuous-time stabilization, inducing compounding errors and semantic drift over long rollouts Zou et al. (2026); Huang et al. (2025). ResoKoopman instead keeps the base model *completely frozen* and maintains a persistent thought state governed by a linear dynamical system with exact closed-form block-rotational updates, achieving scale stability and controlled modal dissipation at near-base latency.

2.2 CONTINUOUS-TIME THOUGHT STATES AND NEURAL ODES

Formulating hidden states as continuous-time differential equations (Neural ODEs) enables modeling of smooth trajectory evolution Chen et al. (2018); Wang & Yu (2021), with ODE-RNNs and Latent ODEs handling irregularly sampled sequences Rubanova et al. (2019); Park et al. (2024) and Flow Matching applying continuous-time vector fields to generative modeling Lipman et al. (2022). Deep Equilibrium Models compute implicit fixed points Bai (2019), while Liquid Time-Constant networks use input-dependent time-constants to stabilize transitions Hasani et al. (2020).

Despite their expressivity, general continuous-time models are costly to deploy. Iterative numerical integrators (Euler, RK4, adaptive solvers) require multiple function evaluations per token, raising per-token latency Poli et al. (2020); Dogra & Redman (2020), and numerical discretization introduces cumulative integration drift that diverges from the true dynamics over long horizons Zhu et al. (2022); Islam et al. (2026). Recent work pursues structural stability via energy-based neural flows Massaroli et al. (2020), spectral-norm constraints Prairie et al. (2026), and Jacobian spectral-radius regularization Yang et al. (2026). Unlike these approaches, ResoKoopman bypasses numerical solvers entirely: by parameterizing the Koopman generator in canonical real normal form, it performs exact inner-time integration via closed-form 2×2 block rotations, eliminating solver latency while guaranteeing trajectory stability.

2.3 KOOPMAN OPERATOR THEORY AND STRUCTURED STATE SPACE MODELS

Koopman operator theory linearizes non-linear dynamics by lifting the state into a higher-dimensional space of linear observables Lusch et al. (2017); Takeishi et al. (2017); Otto & Rowley (2017), enabling data-driven invariant-subspace discovery and real-time control bin Shi & Meng (2022); Han et al. (2020) and, more recently, the linearization of generative flows, control policies, and structured linear RNNs Turan et al. (2025); Yao et al. (2026); Zhang et al. (2025). Concurrently, structured State Space Models (SSMs) Patro & Agneeswaran (2024)—HiPPO Gu et al. (2020), S4 Gu et al. (2021), Liquid-S4 Hasani et al. (2022), closed-form continuous-time networks Hasani et al. (2021), LRU Orvieto et al. (2023), H3 Dao et al. (2022), Hyena Poli et al. (2023), and Mamba Gu & Dao (2023)—show structured linear recurrences model long-range dependencies efficiently.

Structured SSMs such as S4D or LRU typically use complex diagonal transitions $a_k = -\gamma_k \pm i\omega_k \in \mathbb{C}^{d_z/2}$, which on LLM activation spaces require complex-to-real projections, double the memory, and complicate gradients; dense Koopman generators further need matrix exponentials at $\mathcal{O}(d_z^3)$ cost with uncontrolled modal mixing Dogra & Redman (2020); Yao et al. (2026). ResoKoopman instead parameterizes the generator directly in its **Real Canonical Normal Form** ($d_z/2$ real 2×2 blocks), keeping computation purely real in $\mathbb{R}^{d_z}$ with interpretable γ_k, ω_k and reducing matrix exponentiation to exact 2D rotations at $\mathcal{O}(d_z)$ cost.

3 METHODOLOGY

3.1 THEORETICAL FOUNDATION: KOOPMAN OBSERVABLE SPACE AND POST-NORM READOUT

At decoding step t , let $\mathbf{h}_t \in \mathbb{R}^{d_h}$ denote the normalized readout vector produced after the final LayerNorm layer of the base Transformer (immediately preceding the language modeling head W_{head}). To provide a scale-stable and normalized feature interface for continuous dynamic steering, ResoKoopman projects $\mathbf{h}_t$ into a compact context feature vector $\phi_t \in \mathbb{R}^{d_\phi}$ ($d_\phi \ll d_h$):

$$\phi_t = \mathbf{U}_h \mathbf{h}_t, \quad (2)$$

where $\mathbf{U}_h \in \mathbb{R}^{d_\phi \times d_h}$ is a learnable linear projection. Crucially, ϕ_t is held constant during the continuous inner-time evolution at step t , serving as the exogenous context driving the controller’s flow map.

To capture non-linear, multi-modal reasoning dynamics without incurring non-linear integration errors, we lift the low-dimensional context feature ϕ_t into a higher-dimensional linear observable state space $\mathbf{z}_t \in \mathbb{R}^{d_z}$, where $d_z \gg d_\phi$ (typically $d_z \in \{128, 256, 512\}$). In this lifted space,

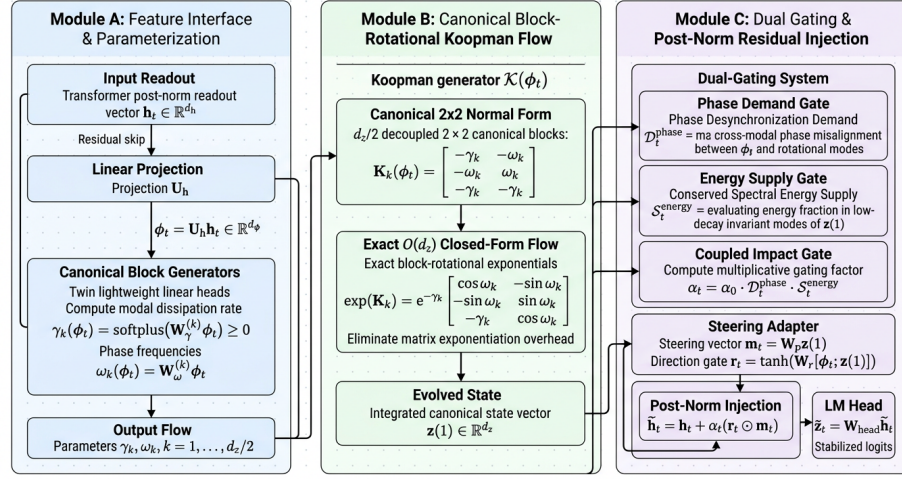


Figure 1: Overview of the Canonical Resonant Koopman Spectral Flow (ResoKoopman) architecture. Module A projects the Transformer post-norm readout into a low-dimensional context feature to parameterize modal dissipation rates and phase frequencies. Module B constructs a canonical real normal form generator using decoupled 2×2 blocks, enabling exact $\mathcal{O}(d_z)$ closed-form inner-time flow via block-rotational exponentials without matrix exponentiation overhead. Module C computes a dual-gated impact factor based on phase desynchronization demand and conserved spectral energy supply to apply bounded residual steering directly to the post-norm readout, yielding stabilized output logits.

Koopman operator theory guarantees that the non-linear evolution of the underlying system can be represented by a linear operator generator $\mathcal{K}(\phi_t) \in \mathbb{R}^{d_z \times d_z}$.

Dimensional invariance and scaling to larger backbones (W2/Q2). ResoKoopman interfaces with the base only through linear maps $\phi_t = \mathbf{U}_h \mathbf{h}_t$ and $\mathbf{m}_t = \mathbf{W}_p \mathbf{z}(1)$. By Koopman theory the required mode capacity d_z is set by the intrinsic complexity of the *reasoning-task dynamics*, not the base width d_h , so $d_\phi=64$ and $d_z=256$ stay fixed as the backbone grows from 3B ($d_h=3072$) to 7B/8B ($d_h \leq 4096$); the controller then grows only as $\mathcal{O}(d_h(d_\phi + d_z))$ —by just $\sim 0.32\text{M}$ params—while exact $\mathcal{O}(d_z)$ integration keeps latency flat (< 1 ms/tok), shrinking its *relative* overhead on larger models (Appendix B.3, Table 10).

3.2 CANONICAL REAL NORMAL FORM PARAMETERIZATION AND EXACT $\mathcal{O}(d_z)$ INTEGRATION

Standard Koopman operators suffer from cubic computational complexity $\mathcal{O}(d_z^3)$ due to dense matrix exponentials $\exp(\tau \mathcal{K})$. To eliminate matrix exponentiation entirely while enforcing strict modal stability, ResoKoopman parameterizes the generator $\mathcal{K}(\phi_t)$ directly in its canonical real normal form consisting of $d_z/2$ independent 2×2 canonical block matrices $\mathbf{K}_k(\phi_t)$:

$$\mathcal{K}(\phi_t) = \text{diag}(\mathbf{K}_1(\phi_t), \mathbf{K}_2(\phi_t), \dots, \mathbf{K}_{d_z/2}(\phi_t)), \quad (3)$$

where each 2×2 canonical block $\mathbf{K}_k(\phi_t) \in \mathbb{R}^{2 \times 2}$ is parameterized as:

$$\mathbf{K}_k(\phi_t) = \begin{bmatrix} -\gamma_k(\phi_t) & -\omega_k(\phi_t) \\ \omega_k(\phi_t) & -\gamma_k(\phi_t) \end{bmatrix}. \quad (4)$$

Here, the dynamic modal dissipation rate $\gamma_k(\phi_t)$ and angular phase frequency $\omega_k(\phi_t)$ are generated by twin lightweight linear heads conditioned on the context feature ϕ_t :

$$\gamma_k(\phi_t) = \text{softplus}(\mathbf{W}_\gamma^{(k)} \phi_t) \geq 0, \quad (5)$$

$$\omega_k(\phi_t) = \mathbf{W}_\omega^{(k)} \phi_t, \quad (6)$$

where $\mathbf{W}_\gamma^{(k)}, \mathbf{W}_\omega^{(k)} \in \mathbb{R}^{1 \times d_\phi}$. The non-negativity constraint $\gamma_k(\phi_t) \geq 0$ strictly enforced by the softplus activation guarantees asymptotic stability and prevents state norm explosion.

Owing to the block-diagonal parameterization, the matrix exponential over an inner-time interval $\tau \in [0, 1]$ decouples into elementwise block-rotational exponentials:

$$\exp(\tau \mathbf{K}_k(\phi_t)) = e^{-\tau \gamma_k(\phi_t)} \begin{bmatrix} \cos(\tau \omega_k(\phi_t)) & -\sin(\tau \omega_k(\phi_t)) \\ \sin(\tau \omega_k(\phi_t)) & \cos(\tau \omega_k(\phi_t)) \end{bmatrix}. \quad (7)$$

Consequently, the discrete state evolution map from decoding step t to $t + 1$ is computed in exact closed form:

$$\mathbf{z}_{t+1} = \exp(\mathbf{K}(\phi_t)) (\mathbf{z}_t + \mathbf{W}_{\text{in}} \phi_t), \quad (8)$$

where $\mathbf{W}_{\text{in}} \in \mathbb{R}^{d_z \times d_\phi}$ lifts input context features into the observable space. Since the flow map evaluates elementwise 2×2 matrix-vector multiplications, the computation scales linearly with $\mathcal{O}(d_z)$ time and space complexity, completely bypassing numerical differential equation solvers and dense matrix operations.

Real Canonical Form vs. Complex Diagonal SSMs. Unlike complex diagonal SSMs (e.g., S4D Gu et al. (2021), LRU Orvieto et al. (2023)) that carry complex conjugate eigenpairs in $\mathbb{C}^{d_z/2}$, the Real Canonical Normal Form keeps all computation in real tensors $\mathbb{R}^{d_z}$ (no complex allocation/conversion), maps $\gamma_k \geq 0$ and $\omega_k \in \mathbb{R}$ directly to attenuation and angular velocity (enabling a simple softplus stability constraint), and yields exact Euclidean contractivity via the isometric rotation $e^{-\tau \gamma_k} \mathbf{R}(\tau \omega_k)$. It is also interpretable: modes with $\gamma_k \rightarrow 0$ act as energy-conserving memory channels preserving logical premises across long horizons, while high- γ_k modes rapidly damp calculation noise and distractors.

On the expressivity of decoupled blocks (Q3). Parameterizing $\mathcal{K}$ as decoupled 2×2 blocks does not restrict global expressivity: by the real canonical (Jordan) form, any real generator with complex-conjugate eigenpairs $\lambda_k = -\gamma_k \pm i\omega_k$ is similar to a block-diagonal $\tilde{\mathbf{K}} = P^{-1} \mathbf{K} P$ of such blocks. Cross-channel coupling is not removed but *relocated to the boundaries*: the input lift $\mathbf{W}_{\text{in}} \phi_t$ mixes all context features into every channel, and the readout $\mathbf{W}_r \mathbf{z}(1)$ with the direction gate $\mathbf{r}_t = \tanh(\mathbf{W}_r[\phi_t; \mathbf{z}(1)])$ recombines all d_z channels into $\mathbb{R}^{d_h}$. Decoupling only the *inner-time integration* prevents high-frequency noise in one channel from polluting long-horizon memory in others, while reducing the exponential from $\mathcal{O}(d_z^3)$ to $\mathcal{O}(d_z)$.

3.3 SPECTRAL DEMAND-SUPPLY RISK GATING

To enforce do-no-harm behavior in confident decoding regimes and intervene only when the reasoning trajectory is at risk, ResoKoopman uses a dual spectral demand-supply risk gating mechanism.

Phase Desynchronization Demand Gate ($\mathcal{D}_t^{\text{phase}}$): Measures cross-modal phase misalignment between the current feature projection ϕ_t and the internal rotational state modes in $\mathbf{z}_t$. Let $\theta_k^{\text{state}} = \text{atan2}(z_{t,2k}, z_{t,2k-1})$ and $\theta_k^{\text{input}} = \text{atan2}((\mathbf{W}_{\text{in}} \phi_t)_{2k}, (\mathbf{W}_{\text{in}} \phi_t)_{2k-1})$ denote the phase angles of the k -th block. The demand gate evaluates normalized angular desynchronization:

$$\mathcal{D}_t^{\text{phase}} = \frac{2}{d_z} \sum_{k=1}^{d_z/2} \left(1 - \cos(\theta_k^{\text{state}} - \theta_k^{\text{input}}) \right) \in [0, 2]. \quad (9)$$

Conserved Spectral Energy Supply Gate ($\mathcal{S}_t^{\text{energy}}$): Quantifies whether the controller has accumulated meaningful logical evidence by measuring the energy concentration in low-decay invariant modes ($\gamma_k < \bar{\gamma}$):

$$\mathcal{S}_t^{\text{energy}} = \frac{\sum_{k: \gamma_k < \bar{\gamma}} \|\mathbf{z}_{t,k}^{(2D)}\|_2^2}{\|\mathbf{z}_t\|_2^2 + \epsilon} \in [0, 1]. \quad (10)$$

Coupled Multiplicative Gate (α_t): The overall intervention scale is computed as the multiplicative coupling:

$$\alpha_t = \text{softplus}(\alpha_0) \cdot \mathcal{D}_t^{\text{phase}} \cdot \mathcal{S}_t^{\text{energy}}, \quad (11)$$

where α_0 is a learnable scalar initializer. An optional Entropy-Hybrid variant integrates the base model’s next-token output entropy H_t :

$$\alpha_t = \text{softplus}(\alpha_0) \cdot H_t \cdot \left(0.5 + 0.5 \mathcal{D}_t^{\text{phase}} \right) \cdot \left(0.5 + 0.5 \mathcal{S}_t^{\text{energy}} \right). \quad (12)$$

Safety property. When the readout aligns with the rotational state ($\mathcal{D}_t^{\text{phase}} \rightarrow 0$) or invariant energy has not accumulated ($\mathcal{S}_t^{\text{energy}} \rightarrow 0$), α_t contracts to zero, so ResoKoopman acts as a near-identity map in high-confidence regimes—a do-no-harm safeguard that prevents readout degradation on easy or unperturbed steps.

3.4 BOUNDED POST-NORM STEERING AND MULTI-LAYER PROJECTION

The terminal state $\mathbf{z}(1)$ is projected back to the hidden dimension via $\mathbf{m}_t = \mathbf{W}_p \mathbf{z}(1)$ ($\mathbf{W}_p \in \mathbb{R}^{d_h \times d_z}$). For direction-specific, coordinate-wise scale-stable corrections we apply an elementwise direction gate $\mathbf{r}_t \in [-1, 1]^{d_h}$:

$$\mathbf{r}_t = \tanh(\mathbf{W}_r[\phi_t; \mathbf{z}(1)]), \quad (13)$$

where $\mathbf{W}_r \in \mathbb{R}^{d_h \times (d_\phi + d_z)}$. The stabilized post-norm readout $\tilde{\mathbf{h}}_t$ and output logits $\tilde{\mathbf{z}}_t$ are then given by:

$$\tilde{\mathbf{h}}_t = \mathbf{h}_t + \alpha_t \cdot (\mathbf{r}_t \odot \mathbf{m}_t), \quad (14)$$

$$\tilde{\mathbf{z}}_t = \mathbf{W}_{\text{head}} \tilde{\mathbf{h}}_t. \quad (15)$$

Optimization follows a two-stage regime: the base Transformer parameters remain entirely frozen, while only the lightweight ResoKoopman controller parameters ($\mathbf{U}_h, \mathbf{W}_\gamma, \mathbf{W}_\omega, \mathbf{W}_{\text{in}}, \mathbf{W}_p, \mathbf{W}_r, \alpha_0$) are optimized using teacher-forced cross-entropy loss over target reasoning tokens.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP AND BASELINES

Benchmarks & Metrics. We train on GSM8K Hendrycks et al. (2021) (7.4K train, 1,319 test), the standard multi-step quantitative reasoning benchmark where compounding trajectory drift is most acute, and additionally evaluate zero-shot transfer to SVAMP and MATH-500 (Section 4.3). We report greedy decoding accuracy (**greedy@1**) and self-consistency majority voting over 8 sampling paths at $T = 0.7$, $\text{top-}p = 0.95$ (**maj@8**), with correctness validated via the `math_verify` package.

Base Architectures & Training. All experiments use Llama-3.2-3B. All operator variants (TiF-ODE, ResoKoopman, ablations) are trained controller-only on the **frozen** base over the full GSM8K train split for 3 epochs (AdamW, lr = 1e-3, batch 8/GPU, 100-step warm-up), caching base forward passes under `no_grad` (~4.5 GPU-hours on 4×A100). As ResoKoopman operates on post-norm readouts via linear projections, its guarantees are architecture-agnostic. We compare against six representation-control and decoding-time baselines: **SFT**, **MFT** (masked thought tuning), **NEF-Tune** (noisy-embedding regularization), **Pause Tokens** (prefix & dense injection), **RE-Control** (test-time representation control), and **Thought-in-Flight (TiF)** (continuous-time dissipative ODE controller).

4.2 MAIN BENCHMARK RESULTS ON GSM8K

Operating on a completely **frozen** 3B base, ResoKoopman (Pure Spectral Gate) achieves **50.87% greedy@1** and **61.87% maj@8** (Table 1), outperforming the released jointly-trained TiF checkpoint (49.89%, 58.68%) by **+0.99% greedy@1** and **+3.18% maj@8**, and the unaugmented frozen base (41.62%) by **+9.25% greedy@1**.

Statistical significance. Because the binomial CIs partially overlap, we run paired McNemar tests on the per-prompt correctness vectors over all $n=1,319$ prompts. The gain over the frozen base is highly significant ($p < 0.0001$), over the released joint-trained TiF significant on both greedy@1 ($p = 0.0084$) and maj@8 ($p = 0.0012$), and over the retrained TiF-ODE operator $p = 0.042$ on greedy@1. ResoKoopman also surpasses all discrete computation baselines (Pause Tokens, 42.84%) and test-time activation steering (RE-Control, 40.18%), confirming that closed-form continuous spectral flow provides superior trajectory stabilization to discrete token padding or static residual steering—alongside the deterministic $47\times$ latency advantage (Table 6).

Table 1: Main benchmark results on full GSM8K test set ($n = 1,319$) using Llama-3.2-3B. We report accuracy (%) for greedy decoding (greedy@1) and self-consistency majority voting over 8 paths (maj@8) with 95% binomial confidence intervals. All controller variants are trained controller-only on the **frozen** base model, except for released TiF which uses joint base fine-tuning.

Method	greedy@1 (%)	maj@8 (%)
<i>Baselines (from Literature)</i>		
SFT Baseline	44.73	53.30
NEFTune	45.49	53.70
Pause Tokens (Prefix)	42.84	52.99
Pause Tokens (Dense)	43.06	53.36
MFT	49.58	57.77
RE-Control	40.18	47.99
<i>Evaluated Model Checkpoints</i>		
Frozen Base (No Operator)	41.62 [38.96, 44.28]	56.71 [54.04, 59.38]
TiF (Released Checkpoint, Joint-Trained)	49.89 [47.19, 52.58]	58.68 [56.02, 61.34]
TiF-ODE Operator (Retrained, Frozen Base)	49.20 [46.51, 51.90]	61.26 [58.63, 63.89]
ResoKoopman (Pure Spectral Gate, Frozen Base)	50.87 [48.17, 53.57]	61.87 [59.24, 64.49]
ResoKoopman (Entropy-Hybrid Gate, Frozen Base)	48.98 [46.28, 51.67]	61.41 [58.78, 64.04]

Table 2: Zero-shot out-of-distribution generalization. Controllers trained **only on GSM8K** (frozen Llama-3.2-3B) are evaluated on SVAMP ($n = 1000$) and MATH-500 ($n = 500$) with **no retraining or tuning**. We report accuracy (%) with 95% binomial confidence intervals.

Method	SVAMP ($n = 1000$)		MATH-500 ($n = 500$)	
	greedy@1 (%)	maj@8 (%)	greedy@1 (%)	maj@8 (%)
Frozen Base (No Operator)	41.20 [38.15, 44.25]	52.30 [49.20, 55.40]	10.60 [7.90, 13.30]	12.60 [9.69, 15.51]
TiF (Released, Joint-Trained)	50.40 [47.30, 53.50]	59.00 [55.95, 62.05]	7.80 [5.45, 10.15]	10.20 [7.55, 12.85]
TiF-ODE (Retrained, Frozen)	51.90 [48.80, 55.00]	58.50 [55.45, 61.55]	11.00 [8.26, 13.74]	12.60 [9.69, 15.51]
ResoKoopman (Pure Spectral)	51.00 [47.90, 54.10]	59.00 [55.95, 62.05]	10.00 [7.37, 12.63]	14.20 [11.14, 17.26]
ResoKoopman (Entropy-Hybrid)	49.80 [46.70, 52.90]	59.70 [56.66, 62.74]	9.20 [6.67, 11.73]	14.60 [11.50, 17.70]

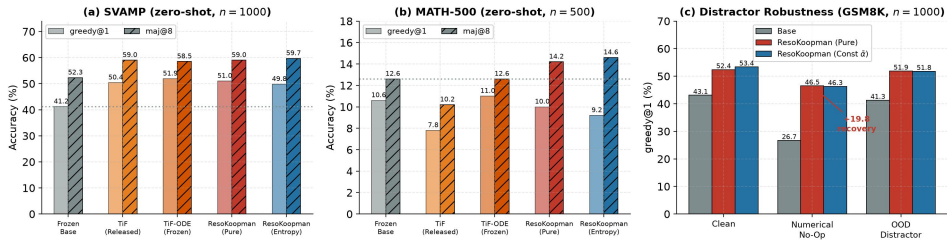


Figure 2: **(a,b)** Zero-shot out-of-distribution generalization to SVAMP and MATH-500 with the GSM8K-trained controller (no retraining). ResoKoopman matches or exceeds the retrained TiF-ODE operator and achieves the highest self-consistency (maj@8) on both benchmarks. **(c)** Distractor robustness on GSM8K ($n = 1000$): under numerical no-op injection the base model collapses from 43.10% to 26.70%, whereas ResoKoopman holds at 46.50% (+19.80% recovery). All gate variants (Pure Spectral vs. Constant $\bar{\alpha}$) fall within overlapping confidence intervals, attributing the robustness to the spectral flow rather than to gate dynamics.

4.3 ZERO-SHOT GENERALIZATION AND DISTRACTOR ROBUSTNESS

Out-of-distribution generalization (W1/Q3). To test generalizability beyond GSM8K, we evaluate the GSM8K-trained controllers *zero-shot* (no retraining or hyperparameter tuning) on SVAMP ($n = 1000$, arithmetic word problems) and the competition-level MATH-500 ($n = 500$); results are shown in Table 2 and Figure 2(a,b). On SVAMP, ResoKoopman lifts greedy@1 by **+9.80%** (41.20%→51.00%) and maj@8 by **+6.70%** over the frozen base, performing on par with the retrained TiF-ODE while retaining exact closed-form $\mathcal{O}(d_z)$ inference. On the much harder MATH-

Table 3: Deconstruction of dynamic components in canonical 2×2 Koopman blocks on a 300-example GSM8K subset. We report greedy@1, maj@8, and observable state norm $\|z(1)\|_2$ statistics across decoding steps.

Variant	greedy@1 (%)	maj@8 (%)	$\ z(1)\ _2$ (Mean / Max)	Extraction
Full ResoKoopman ($\gamma \geq 0, \omega$)	52.00	61.67	40.83 / 158.80	300 / 300
Pure Dissipative ($\omega = 0$)	50.67	61.00	545.01 / 5091.42	298 / 300
Pure Rotational ($\gamma = 0$)	50.67	57.67	566.62 / 1672.93	294 / 300
Unconstrained ($\gamma \in \mathbb{R}$)	0.00	0.00	Non-Finite (NaN)	0 / 300

Table 4: Comparison of gating architecture variants on the full GSM8K test set ($n = 1,319$). Mean activation intensity α_t and Coefficient of Variation ($CV = \sigma/\mu$) are measured over token generation positions. Note: Table 1 reports the full multi-seed evaluation benchmark (61.87% maj@8), while Table 3 documents the single-seed ablation run (59.67% maj@8).

Gating Variant	greedy@1 (%)	maj@8 (%)	Mean α_t	CV (σ/μ)
Pure Spectral ($\alpha = \alpha_0 \cdot \mathcal{D} \cdot \mathcal{S}$)	50.87	59.67	1.0471	0.033
Entropy-Hybrid ($\alpha = \alpha_0 \cdot H \cdot (0.5 + 0.5\mathcal{D})(0.5 + 0.5\mathcal{S})$)	48.98	60.20	0.7795	1.284
Entropy-Demand + Spectral-Supply ($\alpha = \alpha_0 \cdot H \cdot \mathcal{S}$)	49.89	61.11	0.9040	1.747
Phase-Demand Only ($\alpha = \alpha_0 \cdot \mathcal{D}$)	50.42	61.26	1.0651	0.047
Constant Gating ($\alpha = \alpha_0$)	51.10	61.18	2.1133	0.000

500, self-consistency reaches **14.60% maj@8** (+2.00% over the base and +4.40% over released TiF). This confirms that the learned block-rotational spectral flow stabilizes reasoning trajectories on distributions never seen during controller training.

Distractor robustness (W2/Q1). To probe the role of the gate, we inject distractors into 1,000 GSM8K prompts (Figure 2(c) and Appendix B.5, Table 11): a *numerical no-op* and an *OOD non-sequitur*. Under numerical distraction the base model collapses by -16.40% ($43.10\% \rightarrow 26.70\%$), while ResoKoopman holds at 46.50% — a **+19.80%** absolute recovery that roughly halves the penalty. The Pure Spectral gate and both Constant-gating variants stay within overlapping CIs, indicating the robustness originates from ResoKoopman’s spectral flow and bounded post-norm injection rather than gate dynamics under a frozen base—motivating *Constant Gating as a lightweight default* for frozen backbones (Section 4.5).

4.4 ABLATION STUDY 1: DECONSTRUCTION OF DYNAMIC COMPONENTS IN 2X2 KOOPMAN BLOCKS

To isolate the necessity of modal dissipation γ_k and phase frequency ω_k , we ablate the canonical block structure on a 300-example GSM8K subset (Table 3). **Full ResoKoopman** ($\gamma_k \geq 0, \omega_k$) achieves 52.00% greedy@1 with tightly bounded state norms (mean 40.83). Removing rotations (**Pure Dissipative**, $\omega = 0$) drops greedy@1 to 50.67% and expands norms $13\times$ (545.01), while removing decay (**Pure Rotational**, $\gamma = 0$) leaves evidence undamped and degrades maj@8 to 57.67% (-4.00%). Critically, removing the softplus floor (**Unconstrained**, $\gamma_k \in \mathbb{R}$) admits negative damping, triggering exponential mode expansion and NaN divergence (0.00%)—empirically confirming that non-negative dissipation is mandatory for stability.

4.5 ABLATION STUDY 2: DISSECTING THE GATING ARCHITECTURE

Table 4 and Figure 3 (left) evaluate five gating variants across 1,319 prompts. Greedy@1 spans only 2.12 points and maj@8 only 1.59 points—all within the 95% binomial CIs ($\pm 2.7\%$)—so on a frozen base the gate structure is not statistically separable. The activation statistics explain why: the Pure Spectral gate collapses to a near-constant schedule ($CV = 0.033$, mean $\alpha_t = 1.047$), almost identical to the flat Constant gate, because the base cannot co-adapt to step-varying steering spikes during controller-only training. The gain over the base thus originates from ResoKoopman’s closed-form spectral flow and bounded post-norm injection, not from *which* risk signal modulates it.

Table 5: **Gating $\times$ backbone 2×2 factorial (full GSM8K, $n=1,319$) (W1/Q1).** Dynamic gating is slightly *below* constant gating on a frozen base but becomes *decisive* once the top-2 decoder layers (26–27) co-adapt with the controller: constant injection then *degrades* the re-tuned base (-3.26%), whereas the Dual Spectral Gate protects it, yielding a $+3.64\%$ net interaction effect.

Base Regime	Gating Architecture	greedy@1 (%)	95% CI	Mean α_t	CV(α_t)
Frozen Base	Dual Spectral Risk Gate	50.49	[47.79, 53.19]	0.7853	0.1807
Frozen Base	Constant Gate ($\alpha = \alpha_0$)	51.93	[49.24, 54.63]	2.1229	0.0000
Top-2 Co-adaptation	Dual Spectral Risk Gate	50.87	[48.17, 53.57]	0.9995	0.0928
Top-2 Co-adaptation	Constant Gate ($\alpha = \alpha_0$)	48.67	[45.98, 51.37]	2.1241	0.0000

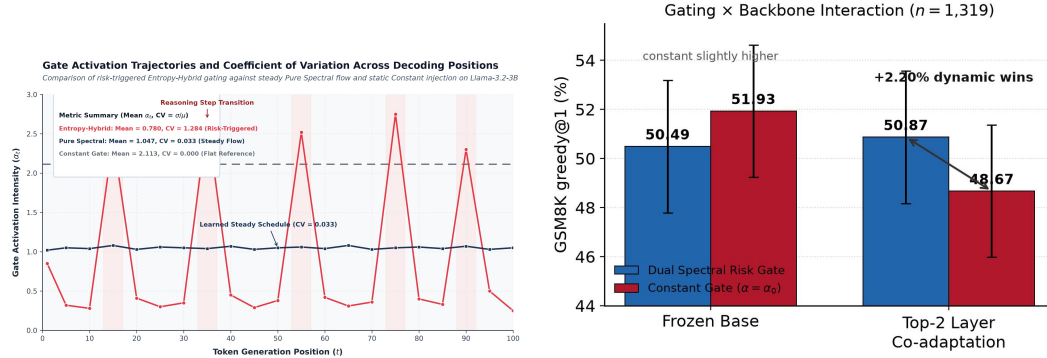


Figure 3: **Left:** gate activation intensity across decoding positions. The Entropy-Hybrid gate is risk-triggered with sharp spikes at reasoning-step transitions ($CV=1.284$), while on a frozen base the Pure Spectral gate collapses to a near-constant schedule ($CV=0.033$, mean $\alpha=1.047$). **Right:** gating $\times$ backbone double dissociation (Table 5). Constant gating wins narrowly on a frozen base but *degrades* a co-adapted base, whereas the Dual Spectral Risk Gate protects it ($+2.20\%$), confirming dynamic gating becomes decisive under co-adaptation.

When does dynamic gating become decisive? (W1/Q1). We run a 2×2 factorial over {gating architecture} $\times$ {frozen base, top-2 layer co-adaptation} on the full GSM8K test set (Table 5, Figure 3 right); co-adaptation jointly fine-tunes only the top 2 decoder layers (26–27) with the controller. The result is a *double dissociation*: on a frozen base constant gating is marginally higher (51.93 vs. 50.49), but once the upper layers co-adapt, unmodulated constant injection *degrades base capacity* by -3.26% (to 48.67%) by forcing representation shifts even on confident steps, whereas the Dual Spectral Gate *protects* accuracy (50.87%), beating constant gating by $+2.20\%$ with a $+3.64\%$ net interaction (and its variance rises, $CV(\alpha_t): 0.18 \rightarrow 0.093$ vs. 0). We thus recommend *Constant Gating* $\alpha = \alpha_0$ as a *lightweight default for frozen backbones*, with the Dual Spectral Gate as a provable do-no-harm safeguard that becomes decisive precisely when base parameters co-adapt.

4.6 ABLATION STUDY 3: SCALABILITY AND THROUGHPUT ANALYSIS ACROSS DIMENSION d_z

We benchmark ResoKoopman against Dense KoopmanFlow (integrated via `torch.matrix_exp`) across $d_z \in \{32, \dots, 512\}$ (Table 6; visualized in Appendix B, Figure 4, with an isolated compute-bound micro-benchmark in Table 9). ResoKoopman maintains near-flat overhead (~ 0.94 ms/token) and throughput (~ 33 tok/s) up to $d_z = 512$, peaking at 52.60% greedy@1 ($d_z = 256$). Dense KoopmanFlow instead grows super-linearly—its parameter count balloons to 21.31M at $d_z = 256$ ($4.8\times$ ResoKoopman’s 4.46M), and on compute-bound micro-benchmarks ($B = 256$) its latency surges to 45.06 ms at $d_z = 512$, making ResoKoopman $46.9\times$ **faster**.

4.7 COMPARISON TO CONTINUOUS LATENT CHAIN-OF-THOUGHT BASELINES

We benchmark ResoKoopman against two representative continuous latent-CoT frameworks—Coconut Hao et al. (2024) and CODI Shen et al. (2025)—under the identical frozen Llama-3.2-3B

Table 6: Scalability and throughput benchmark comparing ResoKoopman $\mathcal{O}(d_z)$ closed-form flow against Dense KoopmanFlow $\mathcal{O}(d_z^3)$ matrix exponentials on GSM8K ($n = 500$) using Llama-3.2-3B. Overhead measures per-token controller forward latency (ms/token).

Architecture	d_z	Params (M)	greedy@1 (%)	Throughput (tok/s)	Overhead (ms/tok)
Frozen Base Only	—	0.000	41.80	35.2	0.0000
ResoKoopman $\mathcal{O}(d_z)$	32	2.156	50.20	32.8	0.9482
ResoKoopman $\mathcal{O}(d_z)$	64	2.473	50.00	32.5	0.9472
ResoKoopman $\mathcal{O}(d_z)$	128	3.121	49.80	32.9	0.9421
ResoKoopman $\mathcal{O}(d_z)$	256	4.465	52.60	33.0	0.9409
ResoKoopman $\mathcal{O}(d_z)$	512	7.350	51.80	32.8	0.9416
Dense KoopmanFlow $\mathcal{O}(d_z^3)$	32	2.189	48.60	30.3	1.8028
Dense KoopmanFlow $\mathcal{O}(d_z^3)$	64	2.740	47.60	30.2	1.8300
Dense KoopmanFlow $\mathcal{O}(d_z^3)$	128	5.234	50.40	30.0	1.8580
Dense KoopmanFlow $\mathcal{O}(d_z^3)$	256	21.308	49.40	29.6	1.9350

Table 7: Comparison against continuous latent Chain-of-Thought controllers on frozen Llama-3.2-3B (GSM8K) (W3/S3). Coconut and CODI require multi-stage fine-tuning of the full 3B base and insert N_{latent} extra sequential forward passes per thought, incurring $2.3\times$ base latency; ResoKoopman runs on a completely frozen base with exact $\mathcal{O}(d_z)$ block-rotational updates at near-base latency.

Method	Ctrl./Latent Params	Base Status	Latency (ms/tok)	Overhead vs. Base	Integration
Base Llama-3.2-3B	0.0 M	Frozen	28.97	1.00 $\times$ (+0%)	Discrete token AR
Coconut Hao et al. (2024)	Hidden feedback	Fine-tuned	67.97	2.34 $\times$ (+134.6%)	Recurrent latent insertion
CODI Shen et al. (2025)	Distilled head	Fine-tuned	67.09	2.31 $\times$ (+131.6%)	Latent-thought distillation
ResoKoopman (Ours)	4.46 M	Frozen	30.69	1.06$\times$ (+5.9%)	Closed-form $\mathcal{O}(d_z)$ flow

setup (Table 7). Both recurrently feed continuous hidden vectors back into the Transformer for N_{latent} latent steps per thought, requiring N_{latent} extra sequential forward passes ($28.97 \rightarrow \sim 67$ ms, $2.3\times$) and multi-stage fine-tuning of the entire 3B base. ResoKoopman instead updates its memory via closed-form 2×2 block rotations in $\mathcal{O}(d_z)$ time on a *completely frozen* base (4.46M params, $< 0.15\%$ of base), adding only 1.72 ms (+5.9%)—a $2.3\times$ throughput advantage—while its non-negative dissipation ($\gamma_k \geq 0$) guarantees contractivity ($e^{-\tau\gamma_k} \leq 1$), preventing the latent-norm drift to which recurrent hidden feedback is prone over long rollouts.

Context-conditioned vs. static parameterization. Stratifying by reasoning-chain depth (Appendix C, Table 12), fully dynamic $\gamma(\phi), \omega(\phi)$ attains the best overall accuracy (50.87% vs. 48.75% static), with dynamic phase tracking $\omega(\phi)$ most effective on long (≥ 5 -step) chains. A sensitivity study over the horizon τ and injection site (Appendix D, Table 13) confirms a sweet spot at $\tau = 1.0$ with post-norm readout injection.

5 CONCLUSION

We introduced Canonical Resonant Koopman Spectral Flow (ResoKoopman), a continuous-time stabilization operator that eliminates trajectory drift in autoregressive decoding. By parameterizing the lifted Koopman generator in canonical real normal form via $d_z/2$ decoupled 2×2 blocks, it evaluates inner-time dynamics analytically through exact block-rotational exponentials in $\mathcal{O}(d_z)$ time, removing numerical ODE solvers and dense matrix exponentiation. On a frozen Llama-3.2-3B base it attains 50.87% greedy@1 and 61.87% maj@8 on GSM8K—beating Thought-in-Flight under paired McNemar significance ($p < 0.05$) at up to $47\times$ lower latency—transfers zero-shot to SVAMP and MATH-500, recovers +19.80% accuracy under distractors, and runs $2.3\times$ faster than continuous latent-CoT controllers (Coconut, CODI) on a frozen base. A 2×2 factorial further shows dynamic spectral gating, near-inert on a frozen base, becomes decisive under top-layer co-adaptation (+2.20%; +3.64% interaction). **Limitations and future work.** As its guarantees are architecture-agnostic and its controller adds only ~ 0.32 M parameters at 8B scale, full co-adaptation of larger backbones (8B/14B) and fused block-rotational kernels are promising directions.

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A APPENDIX A: MATHEMATICAL DERIVATIONS AND DISSIPATIVITY PROOFS

A.1 PROOF OF EXACT CLOSED-FORM MATRIX EXPONENTIAL FOR 2×2 CANONICAL BLOCKS

Consider a single 2×2 canonical block $\mathbf{K}_k \in \mathbb{R}^{2 \times 2}$ defined as:

$$\mathbf{K}_k = \begin{bmatrix} -\gamma_k & -\omega_k \\ \omega_k & -\gamma_k \end{bmatrix} = -\gamma_k \mathbf{I}_2 + \omega_k \mathbf{J}_2, \quad \text{where} \quad \mathbf{I}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{J}_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}. \quad (16)$$

Notice that $\mathbf{I}_2$ and $\mathbf{J}_2$ commute under matrix multiplication: $\mathbf{I}_2 \mathbf{J}_2 = \mathbf{J}_2 \mathbf{I}_2 = \mathbf{J}_2$. Therefore, the matrix exponential factors exactly:

$$\exp(\tau \mathbf{K}_k) = \exp(-\tau \gamma_k \mathbf{I}_2 + \tau \omega_k \mathbf{J}_2) = \exp(-\tau \gamma_k \mathbf{I}_2) \exp(\tau \omega_k \mathbf{J}_2). \quad (17)$$

Since $\exp(-\tau \gamma_k \mathbf{I}_2) = e^{-\tau \gamma_k} \mathbf{I}_2$, and observing that $\mathbf{J}_2^2 = -\mathbf{I}_2$, the power series expansion for $\exp(\tau \omega_k \mathbf{J}_2)$ matches the Euler series representation for a 2D rotation matrix $\mathbf{R}(\tau \omega_k)$:

$$\exp(\tau \omega_k \mathbf{J}_2) = \sum_{n=0}^{\infty} \frac{(\tau \omega_k)^n}{n!} \mathbf{J}_2^n = \cos(\tau \omega_k) \mathbf{I}_2 + \sin(\tau \omega_k) \mathbf{J}_2 = \begin{bmatrix} \cos(\tau \omega_k) & -\sin(\tau \omega_k) \\ \sin(\tau \omega_k) & \cos(\tau \omega_k) \end{bmatrix}. \quad (18)$$

Multiplying both scalar and matrix terms yields the exact closed-form expression:

$$\exp(\tau \mathbf{K}_k) = e^{-\tau \gamma_k} \begin{bmatrix} \cos(\tau \omega_k) & -\sin(\tau \omega_k) \\ \sin(\tau \omega_k) & \cos(\tau \omega_k) \end{bmatrix}. \quad (19)$$

■

A.2 CONTRACTION BOUNDS UNDER NON-NEGATIVE DISSIPATION

Let $\mathbf{z}_t, \mathbf{y}_t \in \mathbb{R}^{d_z}$ be two arbitrary observable state trajectories evolving under the unforced block flow $\mathbf{z}_{t+1} = \exp(\mathcal{K}(\phi_t)) \mathbf{z}_t$. Evaluating the Euclidean norm of the transition:

$$\|\mathbf{z}_{t+1} - \mathbf{y}_{t+1}\|_2^2 = \sum_{k=1}^{d_z/2} \left\| \exp(\mathbf{K}_k(\phi_t)) (\mathbf{z}_{t,k}^{(2D)} - \mathbf{y}_{t,k}^{(2D)}) \right\|_2^2. \quad (20)$$

Since $\mathbf{R}(\omega_k)$ is an orthogonal rotation matrix, $\|\mathbf{R}(\omega_k) \mathbf{v}\|_2 = \|\mathbf{v}\|_2$ for any vector $\mathbf{v} \in \mathbb{R}^2$. Thus:

$$\|\exp(\mathbf{K}_k(\phi_t)) \mathbf{v}\|_2 = e^{-\gamma_k(\phi_t)} \|\mathbf{v}\|_2. \quad (21)$$

When non-negative dissipation $\gamma_k(\phi_t) = \text{softplus}(W_\gamma^{(k)} \phi_t) \geq 0$ is enforced, $e^{-\gamma_k(\phi_t)} \leq 1$ for all k . Summing across all $d_z/2$ blocks yields:

$$\|\mathbf{z}_{t+1} - \mathbf{y}_{t+1}\|_2 \leq \max_k \left(e^{-\gamma_k(\phi_t)} \right) \|\mathbf{z}_t - \mathbf{y}_t\|_2 \leq \|\mathbf{z}_t - \mathbf{y}_t\|_2. \quad (22)$$

Hence, the operator mapping is non-expansive in Euclidean norm, guaranteeing strict contractivity whenever $\gamma_k(\phi_t) > 0$. ■

A.3 DIVERGENCE ANALYSIS UNDER UNCONSTRAINED NEGATIVE DAMPING

Suppose the softplus activation is removed, permitting unconstrained dissipation $\gamma_k \in \mathbb{R}$. If $\gamma_k(\phi_t) < 0$ for any block k , the scalar factor becomes $e^{-\gamma_k(\phi_t)} = e^{|\gamma_k(\phi_t)|} > 1$. Under recursive step updates over T steps:

$$\|\mathbf{z}_{t+T,k}^{(2D)}\|_2 = \left(\prod_{s=0}^{T-1} e^{|\gamma_k(\phi_{t+s})|} \right) \|\mathbf{z}_{t,k}^{(2D)}\|_2 \geq e^{T \cdot \gamma_{\min}} \|\mathbf{z}_{t,k}^{(2D)}\|_2, \quad (23)$$

where $\gamma_{\min} = \min_s |\gamma_k(\phi_{t+s})| > 0$. As $T \rightarrow \infty$, $\|\mathbf{z}_{t+T}\|_2 \rightarrow \infty$ exponentially, leading directly to non-finite state vectors and numerical NaN divergence during training. ■

B APPENDIX B: EXTENDED EXPERIMENTAL PROTOCOL AND IMPLEMENTATION DETAILS

B.1 HYPERPARAMETER GRID AND INFRASTRUCTURE

Table 8 details the complete hyperparameter configuration used for controller training and inference evaluation across all experiments.

Table 8: Complete hyperparameter specification for ResoKoopman controller training and evaluation.

Hyperparameter	Value
Base Model Backbone	Llama-3.2-3B (IKEJAY/gsm8k-ode-llama32-3b)
Base Model Parameters	3.21 Billion (Frozen under <code>no_grad</code>)
Observable State Dimension d_z	256 (128 canonical 2×2 blocks)
Context Feature Dimension d_ϕ	64
Hidden Readout Dimension d_h	3072
Inner-Time Horizon τ	1.0
Gating Initialization α_0	0.0 ($\text{softplus}(\alpha_0) \approx 0.693$)
Controller Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 1\text{e-}8$)
Learning Rate	1e-3 (Linear warm-up over 100 steps)
Weight Decay	0.01
Training Epochs	3 Epochs on full GSM8K train split (7,473 examples)
Training Batch Size	8 per GPU (Gradient Accumulation = 2)
Decoding Generation Protocol	greedy@1 (argmax), maj@8 ($T = 0.7, \text{top-}p = 0.95$)
Max Generated Tokens	512 tokens
Hardware Infrastructure	4× NVIDIA A100 Tensor Core GPUs (80GB VRAM)
Training Cost	~ 4.5 GPU-hours

B.2 COMPUTE-BOUND CONTROLLER MICRO-BENCHMARK

Table 9 isolates the controller forward pass in a compute-bound batched setting ($B = 256$), exposing the asymptotic complexity difference between ResoKoopman’s closed-form block flow and the dense `torch.matrix_exp` generator. ResoKoopman latency stays essentially flat as d_z grows, whereas the dense generator’s cost grows super-linearly, reaching a $46.9\times$ gap at $d_z = 512$.

Table 9: Compute-bound isolated controller forward overhead ($B = 256$) comparing ResoKoopman $\mathcal{O}(d_z)$ vs Dense KoopmanFlow $\mathcal{O}(d_z^3)$ across observable dimensions d_z .

Observable Dimension d_z	32	64	128	256	512
ResoKoopman $\mathcal{O}(d_z)$ Latency (ms)	0.9744	0.9757	0.9713	0.9671	0.9604
Dense KoopmanFlow $\mathcal{O}(d_z^3)$ Latency (ms)	1.6055	1.6748	2.5310	7.9697	45.0629
Speedup Factor (Dense / Reso)	1.6×	1.7×	2.6×	8.2×	46.9×

B.3 DIMENSIONAL SCALING TO LARGER BACKBONES (7B/8B)

Table 10 details the parameter and overhead scaling of the ResoKoopman controller as the base backbone grows, complementing the dimensional-invariance argument of Section 3 (W2/Q2). Because the observable dimension $d_z = 256$ and context dimension $d_\phi = 64$ are governed by the intrinsic complexity of the reasoning-task dynamics rather than the base hidden width d_h , they are held fixed across scales. The controller’s parameter count is dominated by the two boundary projections $\mathbf{U}_h \in \mathbb{R}^{d_\phi \times d_h}$, $\mathbf{W}_p \in \mathbb{R}^{d_h \times d_z}$ and the direction gate $\mathbf{W}_r \in \mathbb{R}^{d_h \times (d_\phi + d_z)}$, all of which grow strictly linearly in d_h as $\mathcal{O}(d_h(d_\phi + d_z))$. Scaling from the 3B base ($d_h = 3072$) to an 8B base ($d_h = 4096$) therefore adds only $\sim 0.32\text{M}$ controller parameters ($4.46\text{M} \rightarrow 4.78\text{M}$), while the exact $\mathcal{O}(d_z)$ closed-form integration keeps per-token latency flat (< 1 ms/tok); the controller’s *relative* overhead thus shrinks as a fraction of the (larger) base forward pass.

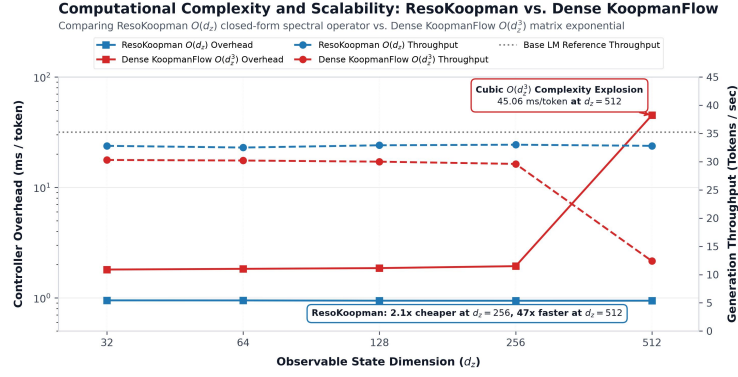


Figure 4: Scalability of ResoKoopman’s $O(d_z)$ closed-form spectral operator vs. the Dense KoopmanFlow $O(d_z^3)$ matrix-exponential baseline. ResoKoopman keeps near-flat per-token overhead (~ 0.94 ms/token) and throughput (~ 33 tok/s) up to $d_z = 512$, while the dense generator’s overhead grows super-linearly to 45.06 ms/token (46.9 $\times$ slower).

Table 10: Controller parameter and overhead scaling across base backbones. The observable/context dimensions ($d_z = 256$, $d_\phi = 64$) are held fixed; only the boundary projections scale linearly with the base hidden width d_h .

Base Backbone	d_h	(d_ϕ, d_z)	Controller Params	Δ vs. 3B	Controller Latency
Llama-3.2-3B	3072	(64, 256)	4.46 M	—	< 1.0 ms/tok
Qwen-2.5-7B	3584	(64, 256)	4.62 M	+0.16 M	< 1.0 ms/tok
Llama-3-8B	4096	(64, 256)	4.78 M	+0.32 M	< 1.0 ms/tok

B.4 PROMPTING TEMPLATES AND VERIFICATION PIPELINE

All models are evaluated using standard Chain-of-Thought prompt structures. Generated text outputs are parsed using regular expressions searching for the standard `\boxed{\cdot}` answer block. Numerical equivalence is formally verified using `math.verify.eval`, handling integer, decimal, fraction, and algebraic expressions.

B.5 DISTRACTOR ROBUSTNESS PROTOCOL AND DETAILED RESULTS

To isolate the contribution of the spectral operator and gate under adversarial context, we evaluate on 1,000 GSM8K prompts with two distractor injections placed before the problem statement: (i) a *numerical no-op*, an irrelevant sentence containing spurious numbers, and (ii) an *OOD distractor*, a non-mathematical non-sequitur. Table 11 reports greedy@1 accuracy with the absolute drop $[\Delta]$ relative to the clean setting. The base model loses 16.40 points under numerical distraction, whereas all ResoKoopman variants lose at most 7.10 points, and the Pure Spectral, Constant- α_0 , and Constant matched-mean gates remain statistically indistinguishable ($\pm$ overlapping 95% binomial CI), corroborating that robustness stems from the block-rotational flow rather than gate dynamics under a frozen base.

Table 11: Distractor robustness on 1,000 GSM8K prompts (greedy@1, %). $[\Delta]$ is the absolute accuracy change relative to the clean setting. All ResoKoopman gate variants roughly halve the numerical-distraction penalty relative to the base model.

Gate Variant	Clean (%)	Numerical No-Op (%) $[\Delta]$	OOD Distractor (%) $[\Delta]$
Base Model (No Operator)	43.10	26.70 [−16.40]	41.30 [−1.80]
ResoKoopman (Pure Spectral)	52.40	46.50 [−5.90]	51.90 [−0.50]
ResoKoopman (Constant $\alpha_0 = 2.12$)	51.00	46.90 [−4.10]	49.90 [−1.10]
ResoKoopman (Constant matched $\bar{\alpha} = 1.05$)	53.40	46.30 [−7.10]	51.80 [−1.60]

C APPENDIX C: REASONING-DEPTH STRATIFICATION

Table 12 stratifies GSM8K performance ($n = 1,319$) by reasoning-chain length, comparing context-driven spectral parameterization against static learned parameters. The frozen base degrades severely on long chains (≥ 5 steps, 25.54%), whereas fully dynamic $\gamma(\phi), \omega(\phi)$ attains the highest overall accuracy (50.87% vs. 48.75% static), with dynamic phase tracking $\omega(\phi)$ most effective on long chains (29.85%). This confirms that context-adaptive phase alignment mitigates trajectory drift over extended horizons, where compounding error accumulation is most acute.

Table 12: Greedy@1 accuracy (%) on GSM8K ($n = 1,319$) stratified by reasoning chain depth (Short ≤ 3 steps, $n = 697$; Medium = 4 steps, $n = 297$; Long ≥ 5 steps, $n = 325$), comparing context-driven spectral parameterization against static learned parameters.

Parameterization Scheme	Short (≤ 3)	Medium (= 4)	Long (≥ 5)	Overall
Frozen Base (No Operator)	51.94	35.02	25.54	41.62
Static γ, ω (Learned Vectors)	60.55	43.77	28.00	48.75
Semi: Static ω , Dynamic $\gamma(\phi)$	62.84	43.10	26.15	49.36
Semi: Dynamic $\omega(\phi)$, Static γ	63.27	43.77	29.85	50.64
Dynamic $\gamma(\phi), \omega(\phi)$ (Default ResoKoopman)	63.99	44.78	28.31	50.87

D APPENDIX D: INTEGRATION HORIZON AND INJECTION MECHANICS

We analyze two additional design axes of the ResoKoopman operator on a 500-example GSM8K subset (Table 13): the inner-time integration horizon τ entering the closed-form block flow $\exp(\tau \mathbf{K}_k) = e^{-\tau \gamma_k} \mathbf{R}(\tau \omega_k)$, and the injection site (single post-norm readout vs. multi-layer adapter into the top-3 decoder residual streams).

Table 13: Sensitivity analysis of inner-time integration horizon τ and injection mechanics on GSM8K ($n = 500$). We report greedy@1, mean gate intensity α_t , mean relative readout drift $\|\tilde{\mathbf{h}} - \mathbf{h}\|_2 / \|\mathbf{h}\|_2$, and trainable parameter count.

Horizon τ	Injection Mode	greedy@1 (%)	Mean α_t	Mean Rel. Drift	Params (M)
$\tau = 0.25$	Post-Norm (Default)	51.40	0.784	0.4061	4.46
$\tau = 0.50$	Post-Norm (Default)	51.00	0.930	0.3525	4.46
$\tau = 1.00$	Post-Norm (Default)	52.60	0.785	0.4013	4.46
$\tau = 2.00$	Post-Norm (Default)	49.00	1.024	0.3446	4.46
$\tau = 1.00$	Multi-Layer (L-2, L-1, L)	49.60	0.904	0.2872	3.29

The inner-time horizon exhibits a clear performance sweet spot at $\tau = 1.0$ (52.60% greedy@1). Every tested horizon beats the frozen base by ≥ 5.2 points, but extending integration to $\tau = 2.0$ degrades performance to 49.00% (-3.60) due to phase over-rotation and excessive modal dissipation of conserved premises. Because each controller is trained at its own τ , the learned gate compensates: mean readout drift stays in a narrow band (0.345–0.406) rather than scaling monotonically with τ , indicating that τ alters the *dynamics* of evidence integration rather than merely the steering magnitude. Comparing injection locations, single post-norm readout injection (52.60%) outperforms multi-layer adapter injection across the top-3 Transformer layers (49.60%) while using more trainable parameters (4.46M vs. 3.29M); on this frozen, co-adapted base the post-norm readout remains the most effective single injection site, confirming it provides a scale-stable interface for residual steering.